

Shadow

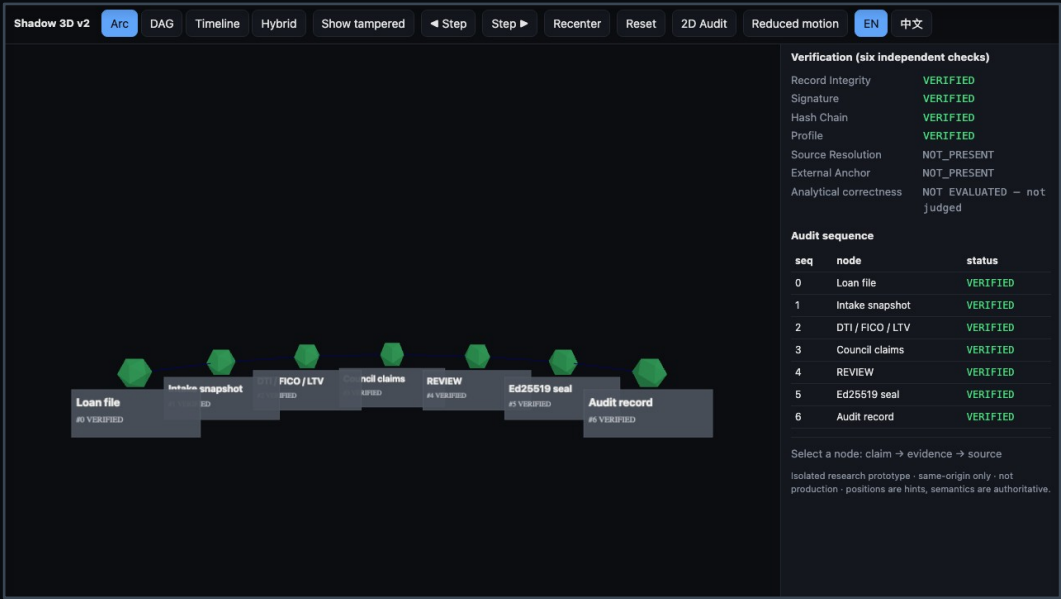
A Cryptographically Verifiable Evidence
and Spatial Audit System for AI Decisions

From AI answers to independently verifiable decision evidence.

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Practice presentation · 2026-07-21



Real render — Shadow 3D audit arc (research prototype)

An answer is not an evidence record

AI ANSWER

“Approve the loan.”

fluent · unverifiable



EVIDENCE RECORD

source → tool action → sequence → signature → verifier

structured · signed · independently checkable

SOURCE

What did it rely on?

ACTION

What did it do?

INTEGRITY

Was it changed after?

INDEPENDENCE

Can someone else verify it?

Ordinary logs are mutable and platform-specific; a model’s explanation is generated prose, not a record.

From more opinions to verifiable evidence

ORALLEXA

Multiple AI perspectives debate an answer



THE LESSON

More voices do not create verifiability

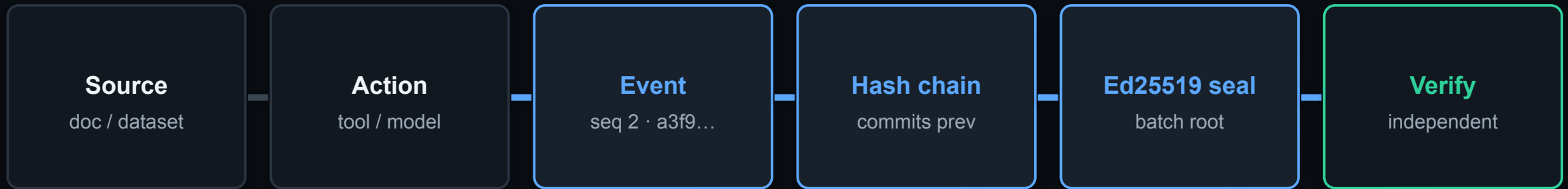


SHADOW

Portable evidence + independent verification

Trust moved **down a layer** — from the voices to the evidence beneath them. A refinement, not a failure.

A cryptographically bound sequence



Change one earlier event → the **first broken link** and its downstream impact become visible.

VERIFIES OFFLINE

SHA-256 + Ed25519

One core, three profiles, many surfaces

EXPERIENCES

CLI · MCP (11) · HTTP · verify.html · Unity Shadow Lens · Three.js replay · explainers

PROFILES

banking-v1 · data-science-v1 · coding-agent-v1

SHADOW CORE

canonical events · hash chain · Ed25519 · source maps · claim–evidence graph · ingest audit (structural)

PORTABLE EVIDENCE BUNDLE

verified by
anyone,
offline

RED
ending

Same verification grammar flows through every layer; interface maturity is labelled, not hidden.

Same machinery across domains

01

Banking

document → risk claim → source
quote

02

Data Science

dataset → model → metric →
selection

03

Coding Agent

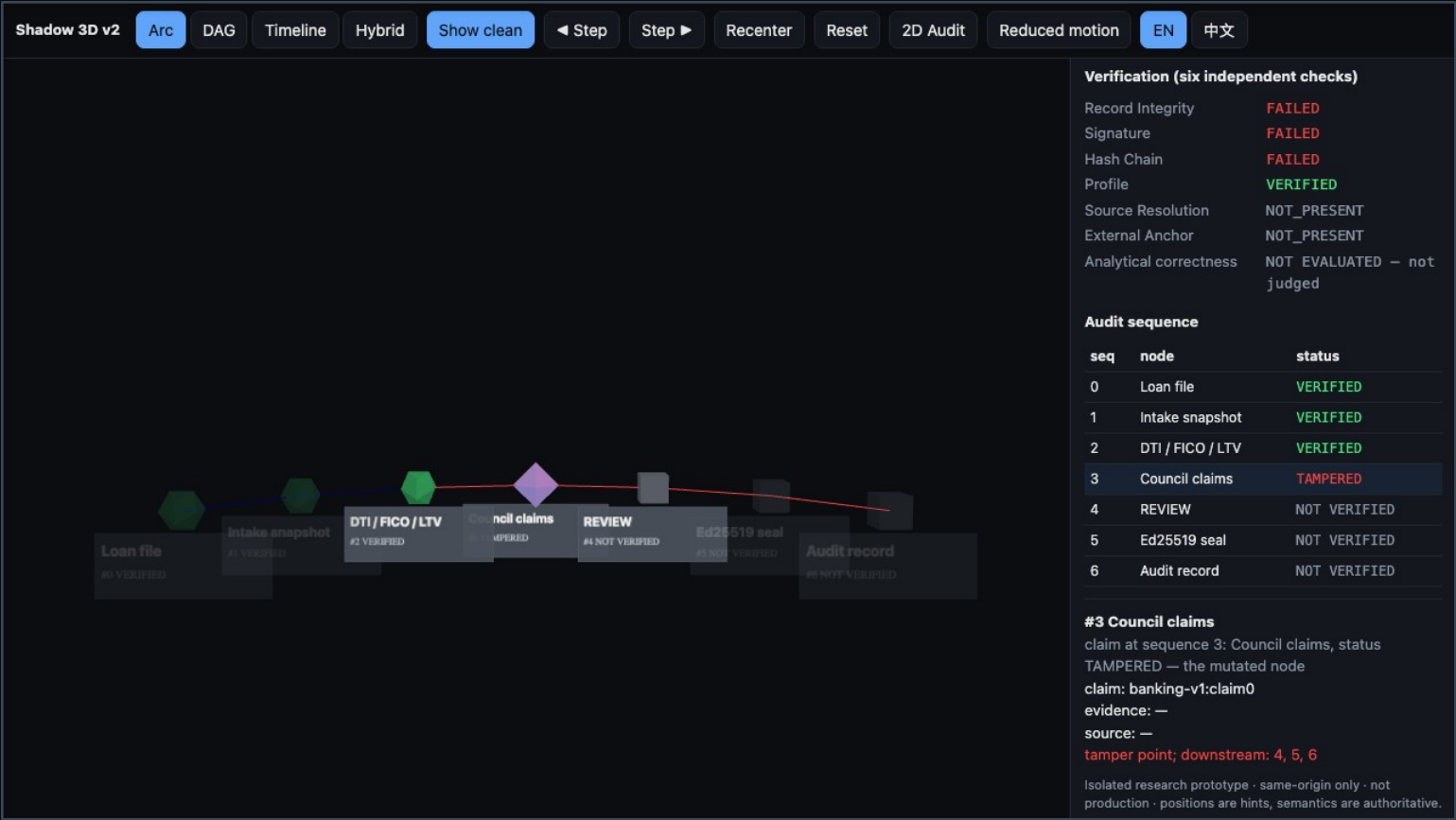
issue → tool call → diff → tests →
commit

Underneath, identical:

sequence → hash chain → signature → source resolution

THE KEY MOMENT

Exact tamper localization



- 1 **Seq 3 — TAMPERED**
the mutated Council-claims node
- 2 **Downstream 4, 5, 6**
NOT VERIFIED — chain invalidated
- 3 **Analytical correctness**
NOT EVALUATED — not judged

Shadow reports:

first failed seq: 3

reason: prev_hash_mismatch

downstream: 4, 5, 6

Integrity failure does not determine analytical correctness — the two stay independent.

Can you trust the verifier page itself?

- 1 **Manifest signature**
VERIFIED
- 2 **Assets match**
SIGNED MANIFEST
- 3 **Independent comparison**
NOT PERFORMED

A page hashing itself is **not independent trust**.

Signing today is a **FIXTURE release key**, not production.

Shadow Verify

Local processing · offline-capable v2.2.0 build 6c3104e EN 中文

VERIFY EVIDENCE

VERIFY THE VERIFIER

Verifier integrity

This page can verify that its loaded assets match a signed release manifest. That does not by itself prove that the manifest or release key was obtained from an independent trusted channel. For strongest assurance, compare the manifest fingerprint with the GitHub Release, npm package, or an approved offline copy.

Load release manifest

Execution mode

STATIC HOST

Release manifest signature

VERIFIED

Release signing-key fingerprint

727d29d3204231f7

Verifier integrity

ASSETS MATCH SIGNED MANIFEST

INDEPENDENT COMPARISON

INDEPENDENT COMPARISON NOT PERFORMED

NOT PERFORMED

Network transparency

External requests (expected: 0)

0

Telemetry

disabled

Uploaded evidence

processed locally — nothing uploaded

Retention by this page

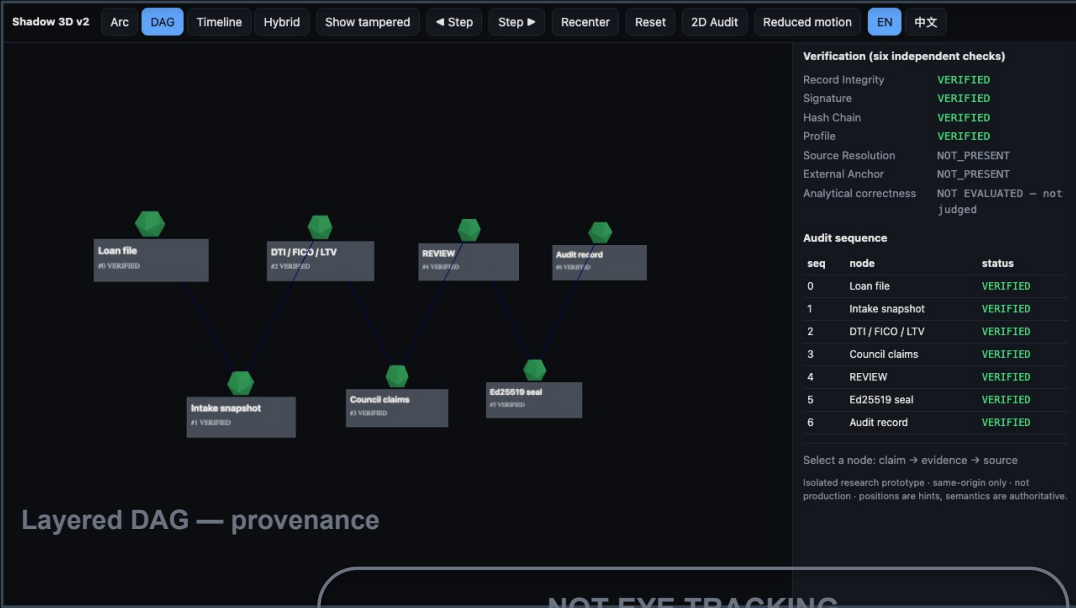
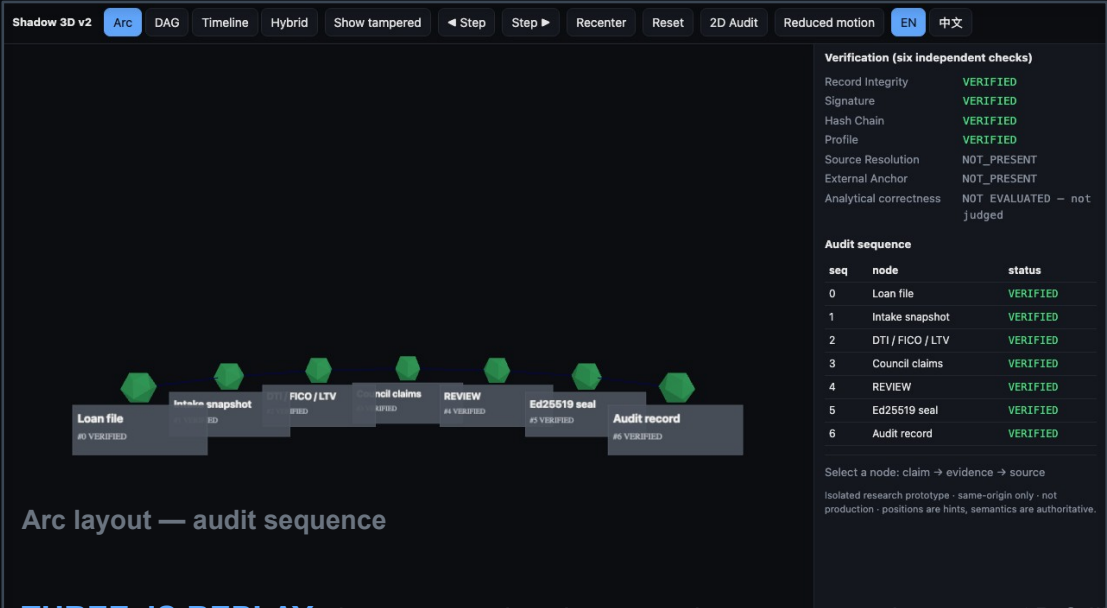
none

Shadow Verify · FIXTURE RELEASE KEY · repo github.com/alex-jb/shadow-mentor

No third-party scripts, no analytics, no CDN fonts. Save locally and it keeps working. 本页无第三方脚本、无分析、无 CDN 字体。

Real render — Verify-the-Verifier (fixture release key)

3D for sequence, provenance, tamper propagation



THREE.JS REPLAY browser-rendered · research prototype · 4 layouts + 2D fallback

UNITY SHADOW LENS unity-authored · Android-built · device validation pending

SHARED 3D SCENE CONTRACT authored + host-tested (shadow-3d-scene-v1)

NOT EYE TRACKING

NO 6DoF ON MOCK

NO USER STUDY YET

Measured today — not inferred

1,858

of 1,861 host tests

0 failed · 3 skipped

re-run 2026-07-21

(V1 said 1,824/1,827)

BROWSER

Chromium 149 · EN + 简体中文 · 0 external requests · 0 CSP violations · offline verify

ANDROID

APK built · 24.4 MB · SHA-256 recorded · NOT device-validated

SPATIAL

Three.js 4 layouts rendered + recorded · scene contract host-tested · no user study

Accurate statuses, not blanket 'pending'

NOW

- ✓ Core evidence + hash-chain verification
- ✓ Claim—evidence graph (host-tested)
- ✓ Bilingual offline verifier
- ✓ Three explainers (audit-chain, reason-code, persona)
- ✓ 3D scene contract — authored + host-tested
- ✓ Three.js 4-layout prototype · Android build

NEXT — CAPSTONE II

- Unity contract integration (pending)
- Beam Pro device validation
- XREAL native input / camera path
- Device performance measurement
- Spatial-vs-2D user study (RQ4)
- Production signing · KMS/HSM · governance

Ingest audit: structural host-tested · semantic production evaluation pending — stated separately, not collapsed.

CONTRIBUTION

Shadow does not ask an auditor to trust the AI's answer.

It gives them evidence they can independently verify.

Sequence. Sources. Signatures. Exact tamper location.

Integrity — not correctness.

Capability ladder — nothing over-claimed

Evidence schema · hash chain · Ed25519

HOST-TESTED

Offline browser verifier (EN + zh-CN)

BROWSER-RENDERED / RECORDED

Verify-the-Verifier manifest

FIXTURE-SIGNED

Claim—evidence graph

HOST-TESTED

3D scene contract (shadow-3d-scene-v1)

AUTHORED + HOST-TESTED

Three.js 4-layout replay

BROWSER-RENDERED · RESEARCH-PROTOTYPE

Unity Shadow Lens

UNITY-AUTHORED · DEVICE-PENDING

Android mock APK

ANDROID-BUILT (not device-validated)

Ingest audit — structural / semantic

HOST-TESTED / SOURCE-AUTHORED

Spatial comprehension benefit (RQ4)

NOT IMPLEMENTED — no user study

What a valid signature does and does not prove

ESTABLISHES

- ✓ The bytes match the sealed record
- ✓ The chain is unbroken (or where it breaks)
- ✓ The batch root was signed by a known key
- ✓ Conformance to a domain profile

DOES NOT ESTABLISH

- ✗ That the cited source was truthful
- ✗ That the conclusion is correct
- ✗ That signing is production-grade
- ✗ That any regulation mandates it

Stated directly

- Evidence is fixture data; limited live-provider validation
- Production signing not implemented (fixture release key)
- Device validation pending (Beam Pro); XREAL native not integrated
- No completed user study — RQ4 spatial benefit unproven
- No proof of source truth; no proof of analytical correctness
- No production KMS/HSM, key rotation, or durable storage
- No full PII-retention framework or incident-response process
- Semantic ingest audit: production evaluation pending

Never depend on a live device

A · Live app

serve verifier package locally

B · Browser package

open the frozen offline package

C · Recorded video

media/.../shadow-verify-full-demo.mp4

D · Slide screenshots

embedded figures — one keystroke away

Decision rule: if the live app doesn't come up cleanly in 20 seconds, drop to the recorded video and keep talking.